GDTR: Layer-wise Settling Depth Reveals Biological Grammar in Genomic Foundation Models

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Abstract

Genomic foundation models encode rich sequence regularities, yet existing tools rarely answer *where* in the layer stack a specific biological grammar becomes stable. We introduce GDTR (*Genomic Deep-Thinking Ratio*), a training-free residual-stream lens that assigns each nucleotide token a *settling depth* $c(t)$: the first layer at which its intermediate representation stabilises against the post-final-norm reference. On Evo 2 7B, splice donor/acceptor sites and ENCODE enhancer-like cCREs settle ~ 2 layers earlier than intronic and coding contexts, and a single chr22 calibration transfers to held-out chr17 (94.6% effect retention). Targeted perturbations separate motif detection from grammar integration: editing the central GT donor motif deepens settling, whereas shuffling the flanking grammar lets the isolated motif settle 3.18 layers earlier. Finally, six ClinVar molecular-consequence classes differ in the layer at which variant-induced residual disruption peaks (Kruskal–Wallis $p = 3.0 \times 10^{-10}$). Together, these results position settling depth as a layer-wise interpretability axis for genomic foundation models, complementing – not replacing – existing variant scorers.

1. Introduction

Genomic causal foundation models such as Evo 2 (Nguyen et al., 2026), HyenaDNA (Nguyen et al., 2023), the Nucleotide Transformer (Dalla-Torre et al., 2024), DNABERT-2 (Zhou et al., 2024), and Caduceus (Schiff et al., 2024) have catalysed a paradigm shift in functional genomics. Yet their internal layer-wise dynamics remain difficult to interpret. Existing tools ask *where* models attend (Abnar & Zuidema, 2020; Avsec et al., 2021), *what* they encode (Bricken et al., 2023; Cunningham et al., 2023), or *how* variants shift representations (Cheng et al., 2023; Jaganathan et al., 2019). A complementary question is largely unmeasured: *where in the layer hierarchy does a biological sequence element become stable?*

We introduce GDTR, a layer-wise readout of residual-stream commitment depth. The Deep-Thinking Ratio of Chen et al. (2026) measures, in NLP transformers, the layer at which intermediate logit-lens distributions (nostalgebraist, 2020; Belrose et al., 2023; Pal et al., 2023) converge within ε of the final layer. GDTR adapts this idea to nucleotide sequences with three changes: a cosine residual-stream lens for small genomic vocabularies; a running-minimum envelope for non-monotone genomic-model trajectories; and an Evo 2-specific post-final-norm reference that handles an idle final block (§2). Fig. 1 summarises the readout.

This short paper makes three contributions. **(C1)** We define the per-token *settling depth* $c(t)$ for genomic causal LMs, including the architectural handling required for Evo 2's idle final block, and validate that a single chr22 calibration transfers to held-out chr17. **(C2)** We show that splice sites and enhancer-like cCREs settle earlier than surrounding genomic contexts, that the splice signal is not explained by next-token entropy, and that targeted motif/flank perturbations cleanly separate motif detection from grammar integration. **(C3)** We show that ClinVar molecular-consequence classes differ in the layer at which a variant most disrupts the residual trajectory, providing a depth probe rather than a clinical scorer. The unifying claim: biological grammar changes not only *what* a genomic foundation model predicts, but also *where* its representation settles.

2. The GDTR Framework

For a nucleotide sequence of length T processed by a causal language model with L layers, we extract the residual-stream activation $h_\ell(t)$ at every layer $\ell \in \{1, \ldots, L\}$ and token position t. We replace the JSD lens of the parent DTR (Chen et al., 2026) with a cosine residual-stream lens

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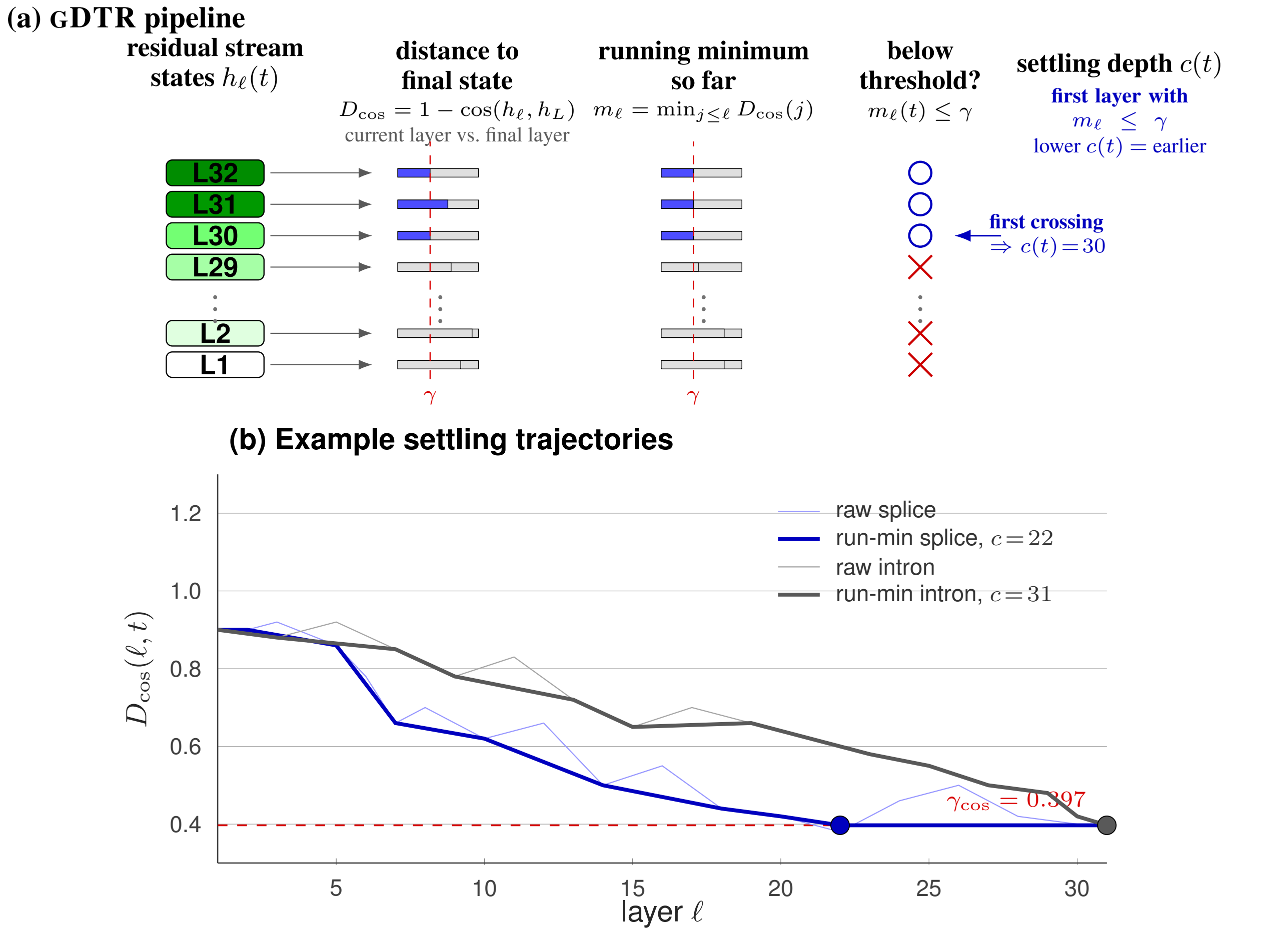

Figure 1. GDTR measures the layer at which a token's representation stabilises. **(a)** Pipeline: each Evo 2 layer emits a residual-stream tap $h_\ell(t)$; the cosine lens compares that tap to the final-layer state $h_L(t)$; a running-minimum envelope tracks the best match so far; and the first threshold crossing defines the settling depth $c(t)$. Lower $c(t)$ means earlier stabilisation. **(b)** Two chr22 example trajectories: a splice-donor token whose running-min crosses $\gamma_{\cos} = 0.397$ at $\ell = 22$ (blue), and an intronic token that settles only at $\ell = 31$ (grey). Faint traces are the raw $D_{\cos}(\ell, t)$; thick traces are the running-minimum envelope (Eq. 2).

that operates directly in hidden-state geometry,

$$D_{\cos}(\ell, t) \;=\; 1 - \cos\!\big(h_\ell(t),\, h_{\text{norm}}(t)\big), \tag{1}$$

where $h_{\text{norm}}(t)$ is the post-final-norm output of the model after the canonical interpretively distinct tap at $L^\star = 29$ has been propagated through the final RMSNorm (blocks indexed $0, \ldots, L-1$ throughout the Evo 2 analysis). We define the *settling depth* via a running-minimum envelope,

$$c(t) \;=\; \min\big\{\, \ell \;:\; \text{run-min } D_{\cos}(\ell, t) \le \gamma_{\cos} \big\}, \tag{2}$$

with run-min $D_{\cos}(\ell, t) = \min_{k \le \ell} D_{\cos}(k, t)$. Lower $c(t)$ means earlier stabilisation; higher $c(t)$ means that the token remains unresolved until later layers. NLP transformers exhibit near-monotonic logit-lens convergence (nostalgebraist, 2020; Belrose et al., 2023); genomic CLMs do not (Hyena alignment rotations, Evo 2 idle final block), so the running-min collapses non-monotonicity into a monotone-non-increasing trace. The threshold $\gamma_{\cos}$ is set by *regional* q_{70} *calibration*: the 70th percentile of the running-min at the penultimate layer over the analysis region. The operating point sits inside a flat 5×5 grid sweep over $(\gamma_{\cos}, \rho)$ (App. A.2).

Evo 2's idle final block. On chr22 sanity sequences, Evo 2's last attention block is architecturally idle: $\max_t |h_{30}(t) - h_{31}(t)| = 0$ exactly (block 31 is a residual passthrough), while $\cos(h_{29}, h_{\text{norm}}) = -0.013$ versus $\cos(h_{30}, h_{\text{norm}}) = \cos(h_{31}, h_{\text{norm}}) = +0.6855$, identifying block 30 as the rotation that aligns the representation with the post-final-norm direction and block 29 as the deepest tap that is still interpretively distinct from h_{norm}. The canonical NLP convention of "tap the last block" therefore breaks; we lock the canonical tap at $L^\star = 29$ and use h_{norm} as the reference (full evidence in App. A.1). This handling is itself a method contribution for subsequent Evo 2 interpretability studies.

Tuned-lens sanity check against frame mismatch. A per-layer tuned-lens affine fit (Belrose et al., 2023) reaches $\geq 98\%$ MSE recovery on $30/32$ Evo 2 layers (canonical $L^{\star} = 29$: 0.9996; App. A.3). This is a sanity check against gross norm-induced frame mismatch: the interior taps are linearly recoverable to the post-norm frame. The framework remains training-free because no affine weights are used at inference.

3. Layer-wise Biological Grammar

3.1. Splice and enhancer annotations settle early

We first test whether $c(t)$ delivers a biologically meaningful, transferable readout at the genome scale. Chromosome 22 serves as the calibration set ($\gamma_{\cos} = 0.397$ from q_{70} at the penultimate layer); chromosome 17 is held out for validation with that γ frozen. The chr17 q_{70} recomputed independently equals 0.394 ($|\Delta| = 0.0023$); the chr17 splice-donor effect reaches 94.6% of the chr22 magnitude (Cohen's $d = -0.349$ vs intron). The calibration transfers without re-tuning.

Context-level ordering. On canonical genomic contexts (Fig. 2a), splice donor and acceptor sites settle ~ 2 layers earlier than the intronic baseline, and the remaining contexts cluster near or above it. Effect sizes are reported as Cohen's d throughout; Mann–Whitney p-values are dominated by the $\sim 10^7$ pooled positions and act only as significance sanity checks. Among the regulatory annotations (Fig. 2b), ENCODE cCRE-ELS enhancer-like elements (Moore et al., 2020) are the clearest non-splice signal at Cohen's $d = -0.190$ against a chr22 per-position background, on a scale where the splice-donor reference reaches -0.433. GTEx eQTLs (The GTEx Consortium, 2020) ($d = -0.022$) and GWAS Catalog SNPs (Sollis et al., 2023) ($d = -0.040$) shift in the same direction but with biologically marginal magnitudes. The $5'$ UTR shift is reported separately: at $5'$ UTR, $c(t)$ couples strongly with next-token entropy ($\rho = +0.41$ vs. $|\rho| \leq 0.15$ elsewhere; §4), so it cannot be compared on the same axis as the entropy-controlled splice and cCRE-ELS signals.

Entropy does not explain the splice signal. A natural concern is that $c(t)$ merely tracks the per-position next-token Shannon entropy H_t of the model. On a control panel of 120 random chr22 windows (720,000 positions) we compute H_t from the post-norm logits and find an overall Spearman $\rho(c, H_t) = -0.079$ (per-context $|\rho| \leq 0.15$ except $5'$ UTR; §4). On the same chr22 panel the splice-donor-versus-intron Cohen's d is -0.452; after regressing c on H_t and re-measuring on the residual it strengthens to -0.583. Next-token entropy therefore does not explain the splice depth ordering.

3.2. Motif edits separate detection from flanking context

Settling depth is not simply a meter of motif intrinsic strength. We dissociate central motif detection from flanking-context integration on $1{,}000$ canonical GT-AG donors (chr22, ± 10 bp pad-averaged c). Real donors settle at $\bar{c} = 26.77$. (i) Replacing the central GT with AA, while preserving the entire ± 100 bp flank, deepens $\bar{c}$ by 0.46 layers (Cohen's $d = -0.09$, paired Wilcoxon $p = 2.3 \times 10^{-32}$), consistent with a small but reliable signal carried by the canonical dinucleotide. (ii) Dinucleotide-shuffling the ± 100 bp flank while preserving the central GT *shallows* $\bar{c}$ by 3.18 layers ($d = +0.51$, $p = 4.1 \times 10^{-59}$): the isolated GT becomes easier to stabilise, whereas the real donor context requires deeper flanking-grammar integration. The within-splice motif breakdown in App. D.2 follows the same cautionary logic: rare or non-canonical motifs can settle earlier than canonical GC-AG, so $c(t)$ should be interpreted as detection plus context integration, not as motif strength alone.

3.3. Variant disruptions peak at consequence-specific layers

For variants, we use the layer at which $|\Delta D_{\cos}|$ peaks as a depth probe: it asks at which layer the residual trajectory changes most between reference and alternate sequence – a depth signature, not a pathogenicity score. We test this on a ClinVar (Landrum et al., 2020) 15-cancer-gene cohort with cached SNV features and a newly forwarded indel panel, class-stratified by ClinVar molecular-consequence (MC) priority, in decreasing order: canonical-splice, nonsense, frameshift, missense, synonymous, intron.

Argmax-layer medians order *intron* $<$ *frameshift* $<$ *nonsense* $<$ *missense* $\approx$ *canonical splice* $<$ *synonymous* (Fig. 3), with the six classes differing at Kruskal–Wallis $p = 3.0 \times 10^{-10}$ (Kruskal & Wallis, 1952). Adjacent-pair Dunn–Bonferroni tests (Mann & Whitney, 1947; Benjamini & Hochberg, 1995) resolve the nonsense $\rightarrow$ missense step ($p_{\text{adj}} < 10^{-3}$), while the remaining adjacent gaps are bounded by per-class sample size ($n = 32$–$1{,}740$). The interpretable result is therefore a population-level depth shift from early-truncating consequences toward missense and splice disruptions.

As a sanity check on trajectory information content, a logistic regression on the full 32-dimensional $\Delta D_{\cos}$ trajectory reaches 10-fold stratified AUROC 0.844 on the $8{,}008$ ClinVar P/LP-vs-B/LB cohort, against 0.729 for the best single-layer tap. Per-gene leave-one-out AUROC stays ≥ 0.77 across the 14 evaluable genes, and DeLong (DeLong et al., 1988) paired comparisons confirm that $\Delta D_{\cos}$ adds significant information beyond Evo 2 log-likelihood ($p = 3.6 \times 10^{-15}$). The full panels, per-layer ablation, and AUROC summary are in App. C (Tables 8, 9; Fig. 6). Com-

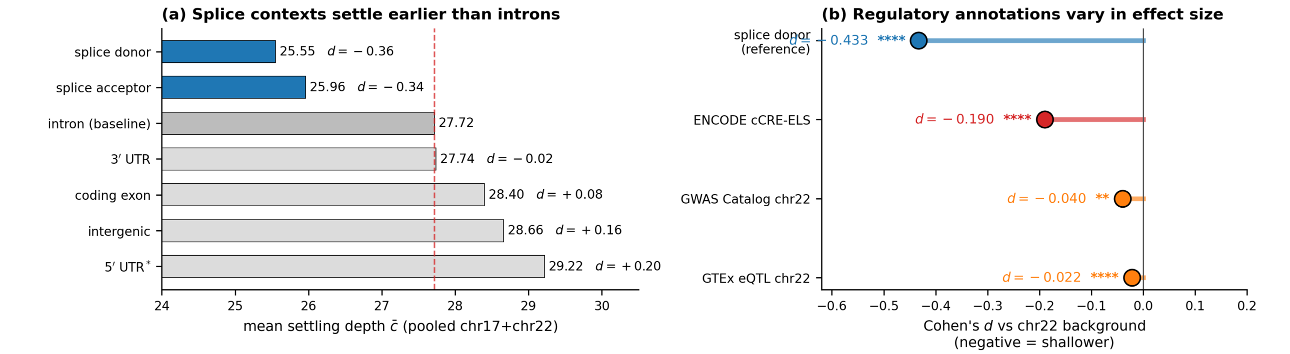

Figure 2. Splice and enhancer annotations shift toward shallower settling. **(a)** Per-context mean settling depth $\bar{c}$ pooled over chr17 (validation, frozen $\gamma_{\text{cos}} = 0.397$) and chr22 (calibration). Splice donor and acceptor (blue) sit ~ 2 layers below the intron baseline (red dashed). The asterisk on "5′ UTR*" marks an entropy-coupled context (§4) reported separately from the entropy-controlled hierarchy. Right of each bar: Cohen's d versus the intron baseline (Mann–Whitney significance: $*, **, ***, ****$ at 0.05, 0.01, 10^{-3}, 10^{-4}). **(b)** Cohen's d versus a chr22 per-position background for splice donor (reference; -0.433), ENCODE cCRE-ELS (-0.190), GTEx eQTL (-0.022) and GWAS Catalog (-0.040). cCRE-ELS gives a measurable enhancer signal; SNP-level eQTL/GWAS shifts are directionally consistent but biologically marginal.

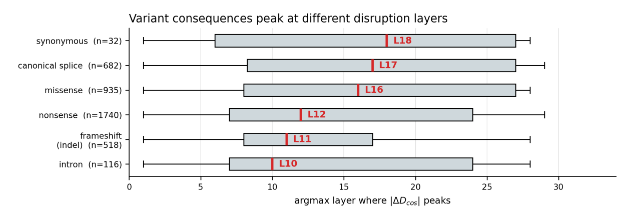

Figure 3. Variant consequences peak at different disruption layers. Argmax layer of $|\Delta D_{\text{cos}}|$ across 4,023 ClinVar P/LP variants in 15 cancer-associated genes, grouped by molecular consequence; class-median layer in red. Kruskal–Wallis 6-way $p = 3.0 \times 10^{-10}$; the largest adjacent jump is nonsense $\to$ missense ($p_{\text{adj}} < 10^{-3}$). The broad overlap emphasises a population-level depth shift, not class separation.

parison against task-specific variant scorers (Cheng et al., 2023; Jaganathan et al., 2019; Rentzsch et al., 2019) is orthogonal to the layer-resolved interpretability question studied here.

3.4. Robustness and tokenisation limits

Extending the chr22 windows to four genomic foundation models (Evo 2 7B, HyenaDNA-large, NT-v2 500M, DNABERT-2) gives a bounded robustness result (Table 1; per-context bars in Fig. 4 and the two-tier rank-concordance summary in Fig. 5, App. B). The donor < intron inequality replicates in the two per-bp causal LMs at depth-normalised scale, and their per-window settling ranks correlate moderately (Spearman $\rho = +0.516$). The two MLMs operate at k-mer/BPE token granularity, so their per-window readout cannot directly test a single-base splice junction. We therefore interpret the cross-model analysis as evidence for replication within per-bp causal LMs and as a diagnosis of tokenisation limits for MLMs.

Table 1. Cross-model summary. Full per-window numbers and MLM tokenisation caveats are in App. B.

Family (model)	Per-pos. readout	Donor < intron
Causal LM (Evo 2, HyenaDNA-large)	yes	yes
MLM (NT-v2, DNABERT-2)	no (per-window)	confounded

4. Discussion and limitations

Four scope conditions frame the interpretation. *(i)* Settling depth is a correlational readout of representational dynamics; establishing causal circuits will require interventional follow-up (e.g. activation patching, sparse-dictionary edits). *(ii)* At 5′ UTR alone, $c(t)$ correlates strongly with prediction entropy ($\rho = +0.41$ vs. $|\rho| \leq 0.15$ elsewhere); the 5′ UTR depth shift therefore reflects a mixture of representational stabilisation and prediction confidence and is reported as a separate, entropy-coupled effect. *(iii)* The current entropy control isolates next-token uncertainty; k-mer rarity, GC content, and dinucleotide composition are natural next composition-matched controls. *(iv)* Single-base splice resolution requires a per-bp causal LM. Adding Caduceus (Schiff et al., 2024) as a second per-position causal model would disentangle architecture from tokenisation in the cross-architecture comparison.

5. Conclusion

GDTR establishes a training-free, layer-resolved interpretability axis for genomic foundation models. Per-token settling depth shows that splice sites and enhancer-like cCREs stabilise earlier than intronic or coding contexts, and the splice signal is not explained by next-token entropy. Targeted motif/flank perturbations show why this depth should be interpreted carefully: the central GT motif contributes a small detection signal, whereas real flanking grammar requires deeper integration. Variant-induced $\Delta D_{\cos}$ trajectories add a second use case, revealing consequence-specific shifts in the layer where residual disruption peaks. Whole-genome scaling, matched-composition controls, and integration with sparse-dictionary (Cunningham et al., 2023) and causal-edit (Meng et al., 2022) methods are left to future work.

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A. Method details

A.1. Architectural quirk handling: Evo 2's idle last block

We discover that Evo 2 7B's last attention block is architecturally idle. Direct verification on chr22 sanity sequences (6 kb $\times$ 100 windows) yields the values in Table 2. Two facts matter: (i) $\max_t |h_{30}(t) - h_{31}(t)| = 0$ exactly, so block 31 is a residual passthrough; (ii) the cosine alignment with the post-final-norm reference h_{norm} jumps from $\cos(h_{29}, h_{\mathrm{norm}}) = -0.013$ (near-orthogonal) to $\cos(h_{30}, h_{\mathrm{norm}}) = \cos(h_{31}, h_{\mathrm{norm}}) = +0.6855$, so block 30 is the rotation that moves the representation into the post-norm direction. The deepest tap that is still *interpretively distinct* from h_{norm} is therefore $L^\star = 29$. The canonical NLP convention of "tap the last block" does not apply to Evo 2; we lock the canonical deep-thinking tap at $L^\star = 29$ and use h_{norm} as the convergence reference.

Table 2. Evidence that Evo 2's last attention block is architecturally idle. Block 31 is bit-identical to block 30 in value but is a distinct tensor; both differ from the post-final-norm reference, identifying block 29 as the deepest interpretively distinct tap.

Quantity	Measured value	Interpretation
$\max\lvert h_{30} - h_{31}\rvert$	0 (exact)	block 31 is residual passthrough
$\texttt{data\_ptr}(h_{30})$ vs. h_{31}	distinct	physically separate copies
$\cos(h_{31}, h_{\mathrm{norm}})$	0.6855	post-norm differs from raw h_{31}
$\cos(h_{30}, h_{\mathrm{norm}})$	0.6855	identical to h_{31}
$\cos(h_{29}, h_{\mathrm{norm}})$	-0.013	deepest distinct tap ($L^\star = 29$)

A.2. Hyperparameter sensitivity

On chr22 the regional q_{70} calibration yields $\gamma_{\cos} = 0.397$. The 5×5 grid sweep (Table 3) produces a wide, flat plateau around the operating point ($\gamma_{\cos} = 0.40,\ \rho = 0.85$) with $\pm 0.10 \times \pm 0.05$ variation. Within this plateau the standard deviation across the 25 cells is 0.06, indicating low sensitivity to small perturbations in either hyperparameter. This robustness was first identified in Phase 0 (HyenaDNA-medium-160k), where the analogous q_{70} calibration produced a best operating point of ($\gamma_{\cos} = 0.50,\ \rho = 0.85$) with Cohen's $d = -1.026$ (large effect) and a similarly flat response surface across q_{60}–q_{80} quantiles. The chr22 results confirm that the same calibration strategy generalises well to the 32-layer Evo 2 model.

Table 3. Effect-size sweep over $(\gamma_{\cos}, \rho)$. Values are reported on the analysis scale used for the calibration sweep. The locked operating point $(0.40, 0.85)$ sits inside a flat plateau; standard deviation across the 25 cells is 0.06.

$\gamma_{\cos} \backslash \rho$	$\rho = 0.70$	$\rho = 0.75$	$\rho = 0.80$	$\rho = 0.85$	$\rho = 0.90$
0.30	5.04	5.18	5.21	5.20	5.15
0.35	5.10	5.21	5.24	5.23	5.18
0.40	5.16	5.25	$\mathbf{5.28}$	$\mathbf{5.26}$	5.21
0.45	5.13	5.22	5.25	5.24	5.19
0.50	5.08	5.17	5.20	5.19	5.14

A.3. Tuned-lens recovery across all 32 layers

Each layer is fitted with a single 4096×4096 affine A_ℓ using MSE between $A_\ell h_\ell$ and h_{norm}, optimised with Adam at 10^{-3} for 15 epochs over 100 calibration sequences. The post-norm tap is the prediction target. Recovery scores at five representative layers are reported in Table 4; $30/32$ layers reach $\geq 98\%$.

Table 4. Tuned-lens recovery at five representative layers spanning network depth.

Layer / block type	Initial MSE	Final MSE (15 epochs)	Recovery
$\ell = 2$ (`hcl`)	1,259	5.6	0.9956
$\ell = 12$ (`hcm`)	742	13.6	0.9816 (worst)
$\ell = 22$ (`hcm`)	418	0.41	0.9990
$\ell = 28$ (`hcs`)	822	0.34	0.9996 (best below tap)
$\ell = 29$ (canonical tap, `hcm`)	510	0.20	0.9996

B. Cross-architecture replication: scope, granularity, two-tier structure

This appendix expands the bounded robustness result of §3.4. We compare four publicly released genomic foundation models that span the per-bp causal-LM and the tokenised MLM design space: Evo 2 7B, HyenaDNA-large-1m, NT-v2 500M, and DNABERT-2 117M. Architectural specifications, the per-window token budget they impose on the same chr22 sequences, end-to-end runtime, and the per-model q_{70}-calibrated $\gamma_{\cos}$ are summarised in Table 5. All four models use the identical chr22 12,978-window panel; numbers are produced by the `results/phase4/per_model_summary.json` pipeline.

Table 5. Architectural specifications, tokenisation, per-window context, runtime, and per-model q_{70}-calibrated $\gamma_{\cos}$ for the four genomic foundation models compared in §3.4.

Model	Architecture	Layers	Hidden	Tokens / 6 kb window	Wall-clock	$\gamma_{q_{70}}$
Evo 2 7B	Hybrid Transformer + StripedHyena 2	32	4096	6,000 (1 bp)	reused	0.396
HyenaDNA-large-1m	Pure Hyena	8	256	6,001 (1 bp + BOS)	$\sim$ 4 min	0.358
NT-v2 500M	Transformer MLM (k-mer)	29	1024	671 ($k = 6$, $\sim$ 4 kb)	$\sim$ 7.5 min	0.533
DNABERT-2 117M	Transformer MLM (BPE)	12	768	$\sim$ 600 (BPE, $\sim$ 3 kb)	$\sim$ 3 min	0.677

Per-context replication. Table 6 reports the per-context mean settling depth for each model. The two per-bp causal LMs (Evo 2, HyenaDNA-large) replicate donor $<$ intron and acceptor $<$ intron at depth-normalised scale ($\sim$ 3 layers below intron in 32-layer Evo 2; $\sim$ 0.34 layers below intron in 8-layer HyenaDNA). The two MLMs tokenise at k-mer/BPE granularity: their per-window readout collapses splice-junction signal into the surrounding window mean and therefore matches the exon-dominant window mean to within rounding, consistent with a tokenisation-induced loss of single-base resolution rather than a model-class disagreement.

Table 6. Per-context mean settling depth across the four models on the identical chr22 12,978-window set. Causal-LM models recover donor/acceptor $<$ intron at depth-normalised scale; MLM models report per-window estimates with no per-position separation.

Model	Kind	L	$\gamma_{q_{70}}$	$\bar{c}$ (intron)	$\bar{c}$ (coding exon)	$\bar{c}$ (donor / acceptor)
Evo 2 7B	per_position	32	0.396	27.84	28.28	25.59 / 25.71
HyenaDNA-large	per_position	8	0.358	6.89	6.67	6.55 / 6.62
NT-v2 500M	per_window	29	0.533	n.a. (intron-dominant 0)	27.80 (exon-dom.)	27.85 (splice-cont.)
DNABERT-2 117M	per_window	12	0.677	n.a. (intron-dominant 0)	11.28 (exon-dom.)	11.27 (splice-cont.)

Pairwise rank concordance. Table 7 shows the pairwise Spearman ρ of per-window mean settling depth across the four models. The pattern is two-tier: within-family correlations are positive ($\rho_{\mathrm{Evo2,Hyena}} = +0.516$; $\rho_{\mathrm{NT,DNABERT}} = +0.663$), whereas every cross-family pair is weakly negative. This is the empirical basis for the "replication within per-bp causal LMs, tokenisation-bound for MLMs" claim of §3.4.

Table 7. Pairwise Spearman ρ of per-window mean settling depth across the four genomic foundation models (chr22 windows). Within-family correlations are positive; cross-family correlations are weakly negative.

	Evo 2 7B	HyenaDNA-large	NT-v2 500M	DNABERT-2
Evo 2 7B	1.000	+0.516	−0.119	−0.188
HyenaDNA-large	+0.516	1.000	−0.287	−0.166
NT-v2 500M	−0.119	−0.287	1.000	+0.663
DNABERT-2	−0.188	−0.166	+0.663	1.000

Visual summaries. The per-context bar chart (Fig. 4) shows the donor/acceptor $<$ intron inequality directly for the per-bp causal LMs and the absence of separation for the tokenised MLMs. Fig. 5 consolidates the rank-concordance heatmap, the depth-normalised donor-vs-intron comparison, and a schematic of the two-tier structure into a single panel.

C. Variant pathogenicity: AUROC summary, per-layer ablation, and panels

This appendix backs up the trajectory-information sanity check in §3.3. The cohort is the 8,008 ClinVar P/LP-vs-B/LB SNVs across 15 cancer-associated genes; the splitter is 10-fold stratified cross-validation with seed 42, and the LOGO-CV

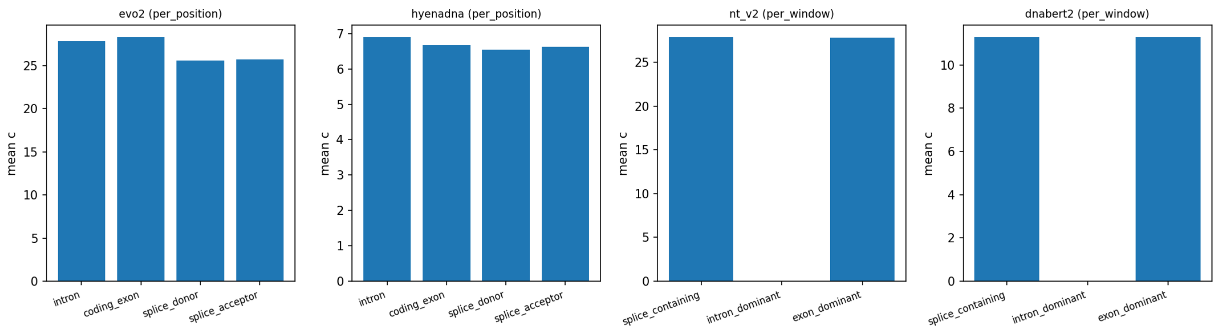

Figure 4. Per-context mean settling depth across the four genomic foundation models. Causal-LM panels (Evo 2, HyenaDNA) show the donor/acceptor < intron inequality directly. MLM panels (NT-v2, DNABERT-2) show no separation at tokenised window resolution. Underlying numbers: `results/phase4/per_model_summary.json`.

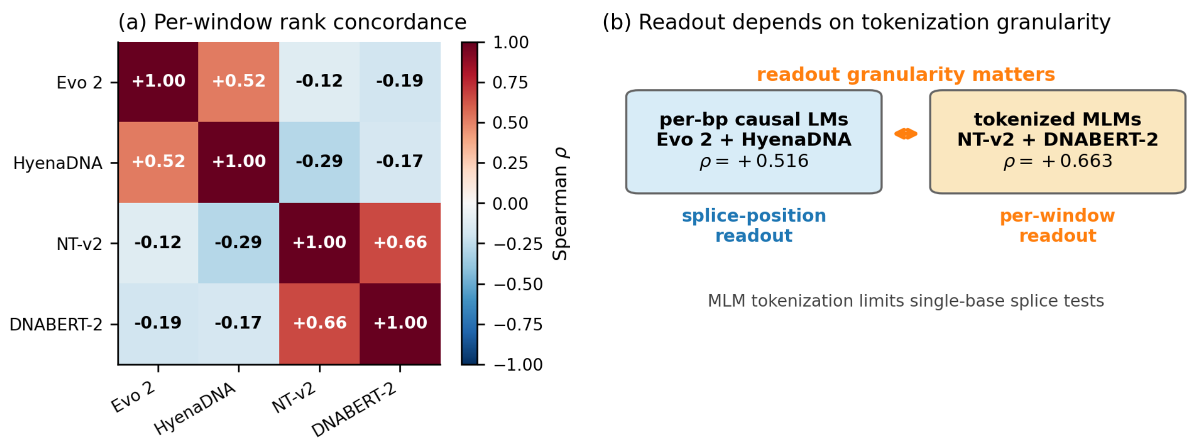

Figure 5. Cross-architecture two-tier structure (numerical detail in Tables 6, 7). **(a)** Pairwise Spearman ρ of per-window mean settling depth across four genomic foundation models on chr22, showing the within-family positive block-diagonal vs. cross-family weak-negative off-diagonal structure. **(b)** Schematic of the two-tier structure with the within/cross-family ρ values, summarising the "per-bp causal LMs replicate, tokenised MLMs are bound by readout granularity" interpretation.

column reports leave-one-gene-out generalisation across the 14 evaluable genes. We treat AUROC strictly as a sanity check on whether the 32-d $\Delta D_{\cos}$ trajectory carries information beyond any single tap; we do not position GDTR as a clinical scorer (§3.3).

Table 8. Feature-level AUROC summary. The primary cosine lens reaches 0.844 with the full 32-d trajectory and only 0.729 at its best single tap – a +0.115 gap. Bracketed numbers are 1,000-bootstrap 95% confidence intervals.

Feature	dim.	Stratified 10-fold AUROC	LOGO-CV AUROC
Best single-layer $\Delta D_{\cos}$ ($\ell = 30$ tap)	1	0.729 [0.717, 0.741]	0.726 [0.694, 0.758]
Best single-layer ΔD_{jsd} ($\ell = 29$ canonical tap)	1	0.794 [0.781, 0.806]	0.787 [0.752, 0.821]
Evo 2 log-likelihood	1	0.751 [0.738, 0.764]	0.793 [0.740, 0.846]
32-d ΔD_{jsd} vector	32	0.823 [0.813, 0.832]	0.821 [0.790, 0.853]
32-d $\Delta D_{\cos}$ vector (primary)	32	**0.844** [0.831, 0.857]	**0.843** [0.811, 0.876]
Ensemble (ΔD+ Evo 2 LL)	33	0.861 [0.851, 0.871]	0.866 [0.832, 0.899]

Per-layer ablation. Table 9 reports single-layer logistic-regression AUROC at representative taps for both lenses. The two lenses agree on the U-shape of layer-wise discriminative mass but disagree on which tap is sharpest: JSD concentrates mass at the canonical tap $\ell = 29$, whereas cosine spreads mass across many taps and accordingly produces a much larger 32-d-vector-vs-best-single-tap gap. The full 32-row table is released as `per_layer_auroc.csv`.

Table 9. Selected per-layer AUROCs. JSD concentrates discriminative mass at the canonical tap $\ell = 29$; cosine spreads it across many taps, producing a much larger 32-d-vs-best-single-tap gap.

Layer	Block type	ΔD_{jsd} AUROC	$\Delta D_{\cos}$ AUROC
0	embed	0.605	0.519
7	`hcs` (shallow)	0.662	0.595
12	`hcm`	0.656	0.555
17	`attn`	0.723	0.646
24	`attn`	0.685	0.612
28	`hcs`	0.565	0.698
29 (canonical tap)	`hcm`	**0.794** (best JSD)	0.604
30 (post-norm−1)	—	0.512	**0.729** (best cos)
31 (idle, see App. A.1)	—	0.499 (degen.)	0.729 ($= h_{30}$)
32-d vector	all	0.823	0.844

ROC, DeLong, and per-gene panels. Fig. 6 provides the four diagnostic panels referenced above. ROC curves (a) confirm the ranking from Table 8; the DeLong paired comparisons (b) show that $\Delta D_{\cos}$ adds significant information beyond Evo 2 ΔLL ($p = 3.6 \times 10^{-15}$); (c) is the per-layer ablation visualised across all 32 taps, and (d) is the leave-one-gene-out AUROC across the 14 evaluable genes (uniformly ≥ 0.77).

D. Splice anatomy beyond the headline contexts

This appendix supports the cautionary message in §3.2: the splice signal is real, but its internal structure is asymmetric and motif-class-dependent.

D.1. Positional fine-profile around donor / acceptor

On chr17 the donor profile reaches its mean settling-depth minimum at position +20 bp ($\bar{c} = 24.06$) and remains ≥ 1.5 layers below intronic baseline ($\bar{c} = 27.69$) out to ±200 bp. On the same chromosome the acceptor profile reaches its minimum at +50 bp ($\bar{c} = 23.33$) and similarly remains depressed beyond ±200 bp. On chr22 both donor and acceptor minima sit at coordinate 0 ($\bar{c} = 23.65$ and 23.64 respectively). The asymmetry — donor minimum exonic-side, acceptor minimum intronic-side — mirrors the asymmetric splice-grammar features (branch point, polypyrimidine tract) that lie predominantly on the intronic side of acceptors.

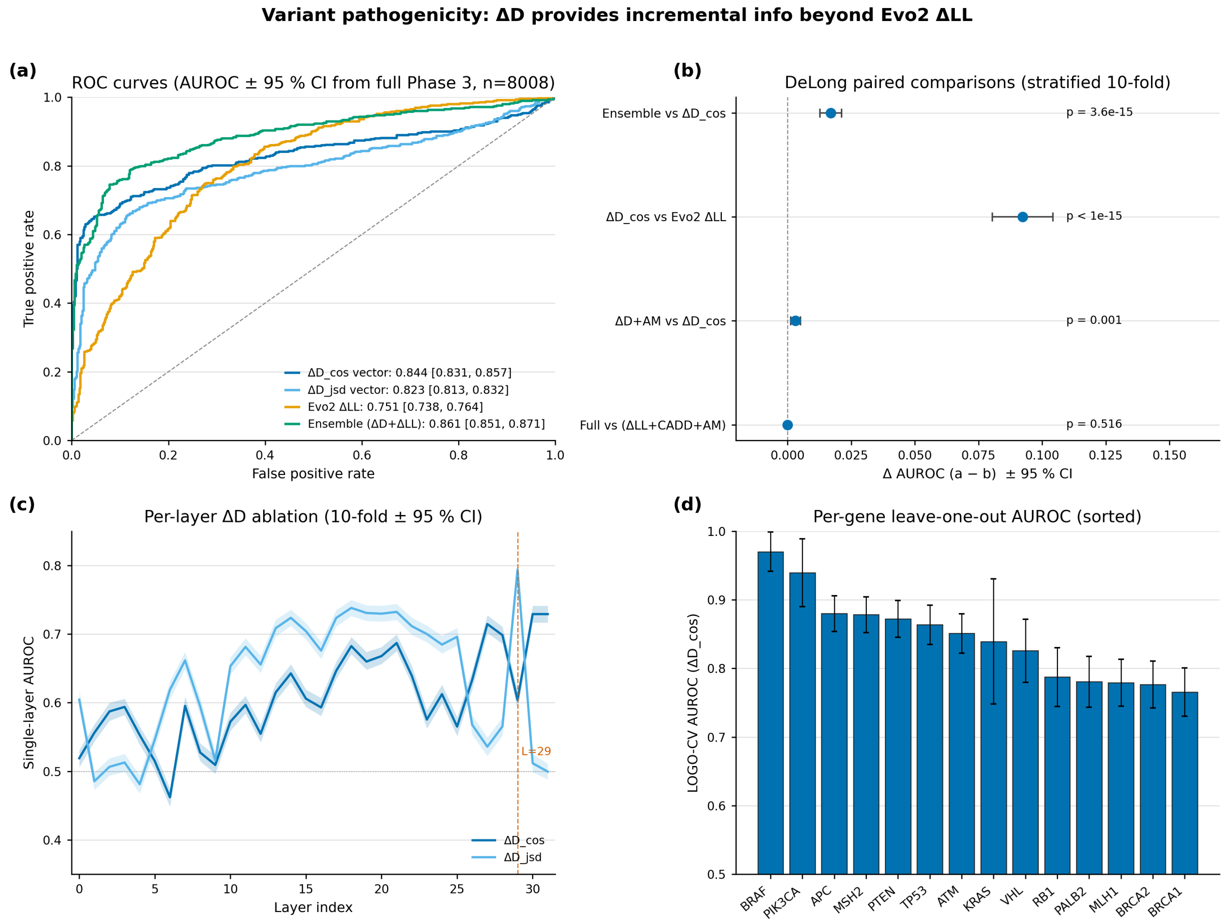

Figure 6. Variant pathogenicity discrimination on 8,008 ClinVar P/LP vs B/LB SNVs across 15 cancer-associated genes (numerical summary in Table 8). **(a)** ROC curves for the four scoring features. **(b)** DeLong paired comparisons: ΔD adds significant information beyond Evo 2 ΔLL ($p = 3.6 \times 10^{-15}$). **(c)** Per-layer single-feature AUROC across all 32 layers – best single tap is $\ell = 29$ for JSD; the 32-d vector beats it by $+0.05$ to $+0.12$. **(d)** Leave-one-gene-out AUROC across 14 evaluable genes is uniformly high (≥ 0.77).

D.2. Canonical vs. non-canonical splice motif breakdown

A finer dissection by motif class – using the genomic dinucleotides immediately downstream of the donor and upstream of the acceptor — reveals an unexpected ordering: *non-canonical* splice donors (AT-AC and other rare motifs, pooled $\bar{c} = 25.13$, $n = 5{,}977$) converge *earlier* than the dominant canonical GT-AG donors ($\bar{c} = 25.79$, $n = 350{,}552$), while the minor canonical GC-AG class is the deepest of the three ($\bar{c} = 27.01$, $n = 5{,}165$). The shallower-than-intron pattern ($\bar{c} = 27.72$) holds for every splice class, but the within-class ordering is the opposite of a naive "stronger motif $\Rightarrow$ shallower recognition" prior. The Cohen's d between canonical GT-AG and non-canonical donors is small (≈ 0.05), consistent with the constrained branch-point and polypyrimidine context that flanks non-canonical introns (Wang & Burge, 2008; Burge et al., 1998); we report the magnitudes honestly rather than over-interpret. The finding replicates qualitatively on the 8-layer HyenaDNA-large at depth-normalised scale.

E. Reproducibility

All experiments use random seed 42 for cross-validation splits and bootstrap resamples. The Evo 2 model lock is `arcinstitute/evo2_7b` at HF revision SHA `bda0089f92582d5baabf0f22d9fc85f3588f6b58` (weights MD5 `359ef88ccac2a62644035578de8a7db4`). Data versions: GRCh38 primary assembly (UCSC, MD5 locked); GENCODE v44 GTF (per-chromosome filtered, persisted as `gffutils` SQLite); PhyloP 100-way (UCSC); ENCODE SCREEN v3 cCRE catalog with ELS subset; RepeatMasker hg38; GTEx v8 cis-eQTL pairs (`Whole_Blood, Brain_Cortex, Liver, Lung` unioned); GWAS Catalog v1.0; ClinVar 2026-04-18. Software stack: `torch 2.4.1+cu124, evo2 0.3.0, vortex 1.0.8, transformer-engine 2.14.0, transformers 4.49.0, scipy 1.13, scikit-learn 1.4`. Hardware: a single NVIDIA H200 (141 GB). Total compute: ~ 20 GPU-hours end-to-end. Code, dataset version locks, and figure-generation scripts will be released at an anonymous URL upon acceptance (an anonymized repository is provided as supplementary material for review).